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Role of angular momentum conservation in unimolecular translational energy release: Validity of the orbiting transition state theory

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The translational kinetic energy release distribution (KERD) for the halogen loss reaction of the bromobenzene and iodobenzene cations has been reinvestigated on the microsecond time scale. Two necessary conditions of validity of the orbiting transition state theory (OTST) for the calculation of kinetic energy release distributions (KERDs) have been formulated. One of them examines the central ion-induced dipole potential approximation. As a second criterion, an adiabatic parameter is derived. The lower the released translational energy and the total angular momentum, the larger the reduced mass, the rotational constant of the molecular fragment, and the polarizability of the released atom, the more valid is the OTST. Only the low-energy dissociation of the iodobenzene ion ($E \approx 0.45$ eV, where E is the internal energy above the reaction threshold) is found to fulfill the criteria of validity of the OTST. The constraints that act on the dissociation dynamics have been studied by the maximum entropy method. Calculations of entropy deficiencies (which measure the deviation from a microcanonical distribution) show that the pair of fragments does not sample the whole of the phase space that is compatible with the mere specification of the internal energy. The major constraint that results from conservation of angular momentum is related to a reduction of the dimensionality of the dynamics of the translational motion to a two-dimensional space. A second and minor constraint that affects the KERD leads to a suppression of small translational releases, i.e., accounts for threshold behavior. At high internal energies, the effects of curvature of the reaction path and of angular momentum conservation are intricately intermeddled and it is not possible to specify the share of each effect. © 2005 American Institute of Physics.
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I. INTRODUCTION

Statistical theories of unimolecular reactions assume that, at a given energy E all possible quantum states of an activated molecule are equally probable. However, in addition to energy, the total angular momentum J of the excited molecule is also a constant of the motion.^{1–4} The corresponding conservation theorems are therefore expected to introduce restrictions to phase-space sampling. What is the efficiency of the corresponding constraints? Furthermore, do additional constraints operate?

The scope of the problem is reduced in the present work. Our contribution is exclusively concerned with the study of the translational energy distribution characterizing unimolecular dissociations taking place in ion beams, i.e., under collision-free conditions.^{1,4–13} Moreover, it is limited to reactions characterized by the absence of any reverse activation barrier, i.e., those for which the potential energy curve increases monotonically along the reaction path. In conventional language, the transition state is then declared “loose.” This often happens in ionic dissociations. Furthermore, we restrict ourselves to reactions where the fragments can be modeled as a spherical or linear top carrying a positive charge interacting with a polarizable atom.

Note that a transition-state calculation of the rate con-

stant is based on the short-range properties of the potential energy surface, i.e., up to the dividing surface that defines the transition state. By contrast, kinetic energy release distributions (KERDs) are mainly determined by the long-range part of the potential. Therefore, a central-field assumption describing the interaction between a point charge and an induced dipole can be expected to provide a reasonable starting point for the study of KERDs. In the various versions of phase-space theory, KERDs are related to recombination reactions via the principle of microscopic reversibility.^{1–4,6–10,14–17} All decay channels are considered to be equally probable provided that energy and angular momentum are conserved. However, the channels may be blocked by centrifugal barriers. A so-called orbiting transition state (OTS) is then associated with the maximum of this barrier.^{1–13,18} Thus, an OTS is an extremely loose transition state whose location depends on the value of the orbital angular momentum. The OTS variant of phase-space theory is hereafter denoted OTST.

The aim of this paper is to analyze the role played by angular momentum conservation in KERDs in the framework of the maximum entropy method (MEM).^{19–30} MEM usually compares the experimental KERD to a prior distribution that takes only energy conservation into account. In the present work, four distributions will be compared using the MEM formalism: the experimental KERD, two possible

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prior distributions, and the KERD predicted by OTST. This approach requires therefore a critical discussion of the validity domain of OTST. The first part of the paper is thus devoted to the derivation of quantitative criteria of validity.

The paper is organized as follows. OTST is briefly reviewed in Sec. II. Section III presents the reactions that will be studied here, both experimentally and theoretically, viz., the dissociation of the bromobenzene and iodobenzene cations, which are regarded as benchmark reactions.^{26,27,31–41} Criteria of validity of OTST are presented in Sec. IV. This theory is shown to be valid only at low internal energies and low translational energy releases. In Sec. V, we present the maximum entropy method^{19–25} as a theoretical instrument that provides a way of determining the constraints that act on the dynamics and of assessing the efficiency of phase-space sampling. Experimental results are presented in Sec. VI and the nature of the dynamical constraints is analyzed. The discrepancies between OTST and experiment are discussed in Sec. VII. The connections between angular momentum conservation and the reduction of the dimensionality of the dynamics from a three-dimensional to a two-dimensional space are also analyzed.

II. THE ORBITING TRANSITION STATE THEORY OF KERDs

A. General formulation

OTST aims at expressing the KERD in terms of calculable energy-level densities when fragments with a vibration-rotational structure interact via a central potential.

Consider a reactant with total angular momentum quantum number J and total energy E that dissociates into a molecular fragment and an atom. In the present paper, the internal energy E is measured with respect to the dissociation asymptote. The fragment is either a linear or a spherical top with rotational quantum number j , rotational constant B , and rotational energy $E_r = Bj(j+1) \approx Bj^2$. The two fragments separate with a relative translational energy ε whose distribution, the KERD, can be expressed as a function of the dissociation rate constant:^{9,10,13}

$$P(\varepsilon|E, J) = k_{\text{diss}}(E, J, \varepsilon) / k_{\text{diss}}(E, J). \quad (2.1)$$

In this equation, $k_{\text{diss}}(E, J, \varepsilon)$ (expressed in units of $\text{energy}^{-1} \text{time}^{-1}$) represents the detailed unimolecular dissociation rate constant at total internal energy E and translational energy ε in the reaction coordinate, while $k_{\text{diss}}(E, J)$ (expressed in units of time^{-1}) is the usual rate coefficient.

The principle of microscopic reversibility^{1–4,13,42} establishes a relationship between the dissociation rate constant and that of the reverse association reaction k_{ass} . The density of vibration-rotational states of the fragments is given by a convolution integral and one has

$$\begin{aligned} \rho_{\text{react}}(E) k_{\text{diss}}(E, J, \varepsilon) &= \int_0^{E-\varepsilon} dE_r \rho_{\text{vib}}(E - \varepsilon - E_r) \\ &\quad \times \rho_{\text{rot}}(E_r) \rho_{\text{transl}}(\varepsilon) k_{\text{ass}}(J, \varepsilon, E_r). \end{aligned} \quad (2.2)$$

In this equation, $\rho_{\text{react}}(E)$ denotes the density of internal

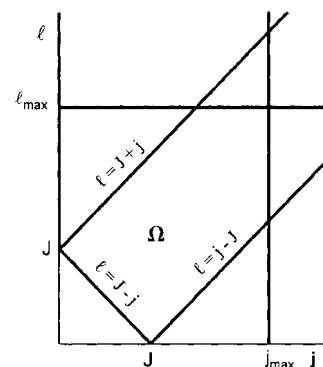


FIG. 1. Graphical representation of the conditions defining the active part Ω of the (j, ℓ) space [Eqs. (2.5)].

states of the reactant, $\rho_{\text{transl}}(\varepsilon)$, $\rho_{\text{rot}}(E_r)$, and $\rho_{\text{vib}}(E - \varepsilon - E_r)$ are, respectively, the density of translational, rotational, and vibrational states of the fragments.

The rate constant of the reverse association reaction $k_{\text{ass}}(J, \varepsilon, E_r)$ is proportional to the capture cross section σ_{capt} .⁴² For fragments of reduced mass μ confined in a box of volume V , we have

$$k_{\text{ass}}(J, \varepsilon, E_r) = \sigma_{\text{capt}}(J, \varepsilon, E_r) (2\varepsilon/\mu)^{1/2} / V. \quad (2.3)$$

If the relative motion of the particles is characterized by an orbital angular quantum number ℓ , the capture cross section is given by^{10,14,15,43}

$$\sigma_{\text{capt}}(J, \varepsilon, E_r) = \left(\frac{\pi \hbar^2}{2\mu\varepsilon} \right) \int_0^\infty (2\ell + 1) P_{\text{capt}}(J, \ell, j, \varepsilon) d\ell. \quad (2.4)$$

The fragments will be captured if the following conditions are fulfilled.^{1,3,4,9,10,13–15,23}

- (1) The angular momentum must be conserved:

$$|J - j| \leq \ell \leq J + j. \quad (2.5a)$$

- (2) The translational energy must be high enough to overcome the centrifugal barrier:

$$\varepsilon \geq V_{\text{eff}}(r_c, \ell), \quad (2.5b)$$

where V_{eff} is the sum of the central potential energy $V(r)$ and of the centrifugal potential energy, and r_c denotes the position of the top of the centrifugal barrier. Equivalently,

$$\ell \leq \ell_{\text{max}}(\varepsilon). \quad (2.5c)$$

These conditions are valid for any central-field potential. They will be explicitly formulated later in the particular case of an ion-induced dipole interaction in Sec. II B. They define an active part Ω in the (j, ℓ) space that is represented in Fig. 1. In that space

$$P_{\text{capt}}(J, \ell, j) = (2J + 1)(2\ell + 1)^{-1}(2j + 1)^{-1}, \quad (2.6)$$

because the coupling between two angular momenta ℓ and j generates $(2\ell + 1)(2j + 1)$ combinations that differ by the value of the projection J_z and correspond to a total angular momentum J in the range $|\ell - j| \leq J \leq \ell + j$. Outside space Ω , $P_{\text{capt}} = 0$.

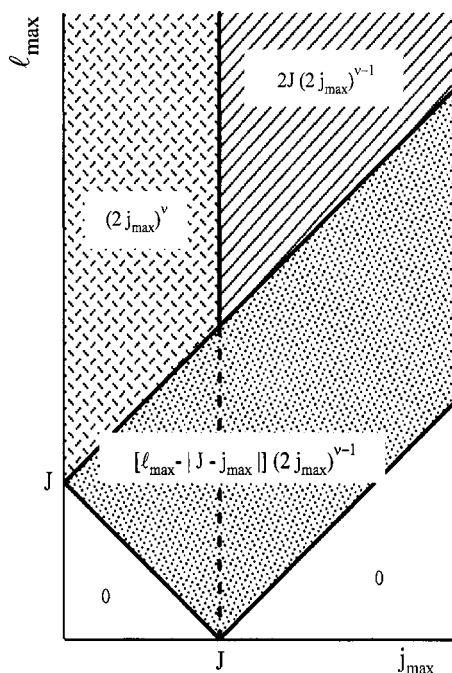


FIG. 2. Graphical representation of the rotational-orbital density of states ρ_{ro} [defined in Eq. (2.8)] multiplied by the rotational constant B , as a function of J , $\ell_{\max}(\varepsilon)$ and $j_{\max} = (E_r/B)^{1/2}$. $\nu=0$, linear rotor; $\nu=1$, spherical top.

Substituting Eqs. (2.2)–(2.6) into Eq. (2.1) and introducing well-known explicit expressions for the densities of translational^{1,23,42} and rotational⁴ states lead to the KERD in the OTST formalism:

$$P_{\text{OTST}}(\varepsilon|E, J) = N(E, J) \int_0^{E-\varepsilon} dE_r \rho_{\text{vib}}(E - \varepsilon - E_r) \times \rho_{ro}(E_r, \varepsilon, J), \quad (2.7)$$

where $N(E, J)$ is a normalization factor, and where the quantity

$$\rho_{ro}(E_r, \varepsilon, J) = \frac{1}{B} \int \int_{\Omega} d\ell dj (2j)^{\nu-1} \delta\left(j - \sqrt{\frac{E_r}{B}}\right) \quad (2.8)$$

is termed the rotational-orbital density of states. $\nu=0$ for a linear rotor and $\nu=1$ for a spherical top.

The integration domain Ω can only be defined piecemeal, as realized by Chesnavich and Bowers.^{6,8} For the spherical top-atom or linear top-atom interactions, the evaluation of ρ_{ro} is simple, and an example is described in Appendix A. The results are summarized in Fig. 2 in terms of two variables: $\ell_{\max}(\varepsilon)$ and $j_{\max} = (E_r/B)^{1/2}$.

Some general features of the KERD already emerge, whatever be the exact shape of the central field potential. Consider first the low-translational energy behavior. Because ε has to be larger than $V_{\text{eff}}(r_c, \ell)$, an E_r -resolved KERD would be sharply truncated at low values of ε . However, the actual distributions are averaged over E_r , then over J , and then over the internal energy E , so that the sharp truncation is replaced by a smooth depletion at low values of ε . Introducing into Eq. (2.7) the expression of ρ_{ro} appropriate for

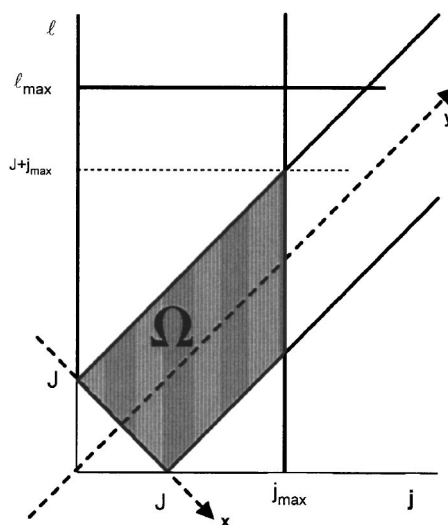


FIG. 3. Definition of the new variables x and y [Eqs. (A1)] that facilitate the calculation of the rotational-orbital density of states ρ_{ro} defined in Eq. (2.8).

low values of ε (or of $\ell_{\max}$), which is derived in Appendix B, and expanding ρ_{vib} around $E_r = BJ^2$ leads to the first-order expression

$$\lim_{\varepsilon \rightarrow 0} P_{\text{OTST}}(\varepsilon|E, J) \propto \ell_{\max}(\varepsilon)^2. \quad (2.9)$$

The function ρ_{ro} is also very simple in the upper part of Fig. 2, i.e., for high values of $\ell_{\max}$ and of ε . This range is defined by the condition $\ell_{\max}(\varepsilon) > J + j_{\max}$. Then, as represented in Fig. 3, the active space Ω does not depend on $\ell_{\max}(\varepsilon)$. Since ρ_{ro} then no longer depends on ε , the dependence of the KERD on ε is directly related to the shape of the function ρ_{vib} . A demonstration that, at high translational energies, P_{OTST} is proportional to the vibrational sum of states is given in Appendix C.

B. Ion-induced dipole interaction

In the case of an ion-induced dipole interaction, an effective potential is generated by adding a quasidiatomic centrifugal term and a rotational term,^{1-4,6,8-10,13,14,23,44-46}

$$V_{\text{eff}}(r) = -\alpha q^2/2r^4 + \ell^2 \hbar^2/(2\mu r^2) + B j^2, \quad (2.10)$$

where α is the polarizability of the noncharged partner and q is the ionic charge.

At a particular value of ℓ , the distance r_c at which the effective potential $V_{\text{eff}}(r, \ell)$ has its maximum is equal to

$$r_c(\ell) = (2\mu\alpha q^2)^{1/2}/\ell \hbar. \quad (2.11)$$

The orbiting energy at the transition state is assumed to be adiabatically converted into translational energy. A system that releases a kinetic energy ε cannot then have an orbital angular momentum higher than^{1,3,4,6,8,10,13,14}

$$\ell_{\max}(\varepsilon) = (8\mu^2\alpha q^2\varepsilon)^{1/4}/\hbar. \quad (2.12)$$

The shortest value of r_c that is involved in the dynamics leading to the release of a translational energy ε is equal to

$$r_c(\varepsilon) = (\alpha q^2/2\varepsilon)^{1/4}. \quad (2.13)$$

Equations (2.9) and (2.12) lead to an $\varepsilon^{1/2}$ threshold law

$$\lim_{\varepsilon \rightarrow 0} P_{\text{OTST}}(\varepsilon|E, J) \propto \ell_{\text{max}}(\varepsilon)^2 \propto \varepsilon^{1/2}. \quad (2.14)$$

III. THE HALOGENOBENZENE CATIONS

The two unimolecular reactions chosen for the present work concern the loss of a halogen atom from an ionized halogenobenzene cation



(with $\text{X}=\text{Br}$ or I). Both have previously received extensive attention,^{26,27,31–41} because they are thought to be a paradigm of a simple bond cleavage characterized by a totally loose transition state.

A. Experiment

The metastable dissociation of these ions has been studied in a forward geometry two-sector mass spectrometer where the electrostatic analyzer (ESA) is followed by a magnet (B). This setup has already been described²⁷ and is reported here only briefly. The parent $\text{C}_6\text{H}_5\text{X}^+$ ions are generated by electron-impact ionization and then accelerated by a voltage difference V_{acc} . Scanning V_{acc} at selected ESA and B settings, one records the mass spectrum of parent ions dissociating to a given fragment ion in the first field-free region, located between the source exit slit and the entrance of the ESA.^{47–49} To get the mass spectrum for fragmentations inside the second field-free region (between the electric and magnetic fields), a scan of the magnetic field is performed.⁵⁰ In these mass spectra, measured in the laboratory reference frame, the release of kinetic energy on the fragment causes a broadening of the signal. Therefore, switching to a center of mass reference frame by a differentiation procedure, the shape of the broadened peak in the mass spectrum gives the translational energy release distribution.^{51–53} The smallness of the translational energy releases in dissociations (3.1) precludes the appearance of angular discrimination and makes the use of more sophisticated data handling procedures^{54–56} unnecessary.

The detected fragment ions are produced within a given time window, corresponding to the transit time of the parent ions in the field-free region. Denoting the entry and exit times of the field-free region as τ_1 and τ_2 , the internal energy distribution of the dissociating ions is the transmission function $T(E)$,

$$T(E) = A\{\exp[-k(E)\tau_1] - \exp[-k(E)\tau_2]\}, \quad (3.2)$$

where A is a normalization constant and $k(E)$ is the dissociation rate constant. Therefore, to work at low internal energy, one must increase the time spent inside the spectrometer by parent ions before dissociation. This can be done by using the second field-free region instead of the first one or by choosing a low accelerating voltage V_{acc} to reduce the velocity of the ions.

The total angular momentum distribution $P(J)$ is simply provided by a Boltzmann distribution at the spectrometer source temperature ($T \sim 150^\circ\text{C}$),

TABLE I. Numerical values (in angstroms) of r_1 [Eq. (4.1)] and of r_2 [Eq. (4.2)] to estimate the validity of the ion-induced dipole approximation. r_{eq} is the equilibrium distance between the halogen atom and the center of the phenyl ring in the parent ion. $\langle E \rangle$ denotes the average internal energy of the parent ion. The average translational energy releases $\langle \varepsilon \rangle$ are experimental measurements.

	$\langle E \rangle$ (eV)	$\langle \varepsilon \rangle$ (eV)	r_{eq}	r_1	r_2 ($J=1$)	r_2 ($J=80$)	r_2 ($J=160$)
$\text{C}_6\text{H}_5\text{H}^+$	1.30	0.20	2.5	1.7	0.39	0.27	0.20
$\text{C}_6\text{H}_5\text{Cl}^+$	1.26	0.091	3.1	2.8	5.1	3.1	2.2
$\text{C}_6\text{H}_5\text{Br}^+$	0.82	0.060	3.2	3.3	9.5	5.2	3.5
$\text{C}_6\text{H}_5\text{I}^+$	0.61	0.051	3.4	3.9	15	7.7	5.2
$\text{C}_6\text{H}_5\text{I}^+$	0.45	0.046	3.4	4.0	15	7.9	5.3

$$P(J) = (2J+1)^{J+1} \exp[-J(J+1)B_0/k_B T], \quad (3.3)$$

where B_0 is the rotational constant of the parent $\text{C}_6\text{H}_5\text{X}$ molecule. For reactions (3.1) this distribution has its maximum near $J \approx 40$ for $\text{X}=\text{H}$ and $J=80$ for $\text{X}=\text{C}$, Br , and I . The distribution ends up at $J \approx 100$ for $\text{X}=\text{H}$ and at $J \approx 160$ for halogens losses.

Some measurements were previously done by Urbain *et al.*^{26,27} Nevertheless, because of the OTST conditions of validity (see Sec. IV, Tables I and II), we were this time especially interested in the iodine loss at low energy. Therefore, to test the reproducibility, we recorded again the spectrum about the iodine loss in the second field-free region with $V_{\text{acc}}=8$ kV. The new KERD is found to be identical with the previous measurement,²⁷ indicating an excellent reproducibility.

B. Modelization

To model these reactions, the vibrational density of states ρ_v of the phenyl ring has been generated by a direct count Beyer–Swinehart algorithm.^{1,2} The vibrational frequencies have been calculated⁵⁷ *ab initio* at the B3LYP/6-31G(d) level recommended by Scott and Radom and scaled according to their prescriptions.⁵⁸ The agreement with the set of frequencies previously calculated by Klippenstein is excellent.⁴¹ Values of 0.226, 0.180, and 0.100 cm^{-1} have been calculated for the three rotational constants of the phenyl ring. However, because our OTST equations have been derived for a spherical top, these values have been replaced by their geometrical mean. For the polarizabilities of the bro-

TABLE II. Adiabatic parameter ξ [defined in Eqs. (4.8) and (4.9)] for the dissociation of the benzene and halogenobenzene ions studied on the microsecond time scale. Adiabatic channel potential curves are meaningful if $\xi > 1$. $\langle E \rangle$ denotes the average internal energy of the parent ion. $\langle \varepsilon \rangle$ is the measured average translational energy released.

	$\langle E \rangle$ (eV)	$\langle \varepsilon \rangle$ (eV)	ξ ($J=0$)	ξ ($J=80$)	ξ ($J=160$)
$\text{C}_6\text{H}_5\text{H}^+$	1.30	0.20	0.04	0.009	0.006
$\text{C}_6\text{H}_5\text{Cl}^+$	1.26	0.091	2.5	0.6	0.4
$\text{C}_6\text{H}_5\text{Br}^+$	0.82	0.060	5.9	1.4	0.9
$\text{C}_6\text{H}_5\text{I}^+$	0.61	0.051	10.0	2.4	1.5
$\text{C}_6\text{H}_5\text{I}^+$	0.45	0.046	10.5	2.5	1.6

mine and iodine atoms, values of 3.05 and 5.04 Å³ were adopted, respectively. The KERDs generated by the OTST equation (2.7) have been calculated in a large range of J values. To be comparable to experimental data, they have been afterwards averaged over the distribution $P(J)$ given by Eq. (3.3).

IV. VALIDITY OF OTST

A number of theories have been applied to ion-molecule reactions, seen either as recombination reactions or as unimolecular dissociations. The statistical adiabatic channel model (SACM) provides the most general framework for their comparison.^{44–46,59} These methods are based on several assumptions.

(1) The potential energy surface increases monotonically along the reaction path and shows no activation barrier other than centrifugal for the association of the fragments.

(2) It is assumed to be properly represented by a multipole expansion. Moreover, the internal energy is assumed to be low enough for the dynamics to be dominated by the long-range part of the interaction potential. Contributions from the higher terms in the multipole expansion are often neglected. Furthermore, the bath of oscillators is assumed to be neatly separated from the other degrees of freedom, which implies that the long-range forces are insensitive to the vibrational motion of the interacting fragments during capture or fragmentation.

(3) The moieties are treated as rigid rotors in formulating the expression for the kinetic energy.

(4) The orbiting energy at the transition state is assumed to be adiabatically converted into fragment translational energy. In other words, ℓ and j are assumed to be good quantum numbers in the region extending from the maxima of the effective adiabatic channel potential curves to an infinite distance between the fragments. Product energy distributions (in particular, KERDs) will be reliably predicted if, after overcoming the centrifugal barrier, the fragments separate adiabatically with respect to all degrees of freedom.

The analysis of KERDs has often been said to provide a decisive argument in favor of statistical theories.^{1,9–13,40} However, what we are looking for is a nonempirical assessment of the range of validity of these theories. For that purpose, we now examine the reliability of assumptions (2) and (4) in the particular case of the dissociation of the halobenzene ions. Up to what energy of the relative motion of the fragments does the model remain valid?

A. Validity of the ion-induced dipole approximation

The KERD can be expected to be reliably calculated only if the distance r_c at which the effective potential $V_{\text{eff}}(r)$ has its maximum is large enough to be located in a region where Eq. (2.10) is valid, i.e., where the assumption of an isotropic ion-induced dipole potential is reliable. Equation (2.11) shows that large values of ℓ are a cause of concern.

The highest possible value of ℓ is the lesser of two quantities: either $\ell_{\text{max}}(\varepsilon)$ given by Eq. (2.12) (as represented in Fig. 1) or $J + j_{\text{max}} = J + (E_r/B)^{1/2}$ (as in Fig. 3).

Consider first the former case, which applies especially when α and μ are small, i.e., when the released atom is a hydrogen.³ The condition $\ell < \ell_{\text{max}}$ amounts to limiting the active part of phase space by the requirement that the fragments must clear the centrifugal barrier [Eq. (2.5b)]. Then, the shortest value of r_c is given by Eq. (2.13). Note that KERDs decrease rapidly as ε increases. Their range is much smaller than the available internal energy because the remainder is absorbed by the bath of vibrational degrees of freedom. In practice, KERDs are observed to become negligibly small when $\varepsilon \geq 3\langle\varepsilon\rangle$, where $\langle\varepsilon\rangle$ is the first moment of the distribution. In that situation, the smallest possible value of r_c of interest is derived from Eq. (2.13):

$$r_1 = (\alpha q^2/6\langle\varepsilon\rangle)^{1/4}. \quad (4.1)$$

In the second case, the active part of phase space Ω is limited by the requirement for angular momentum conservation [Eq. (2.5a)]. This situation is encountered when the released atom is heavy and strongly polarizable, like Br or I. Then, the highest value of ℓ is limited to $[J + (E_r/B)^{1/2}]$. Following an argument developed by Klots,^{1,60} it is reasonable to assume that the bath absorbs large values of both translational and rotational energies with the same efficiency. Hence, invoking the equipartition theorem, the rotational energy E_r present in the system can be expected to range between the same values as translation, i.e., between 0 and $3\langle\varepsilon\rangle$. Therefore, a reasonable order of magnitude for the highest value of ℓ is $[J + (3\langle\varepsilon\rangle/B)^{1/2}]$. The smallest value of r_c of interest is then derived from Eq. (2.11):

$$r_2 = \frac{(2\mu\alpha q^2)^{1/2}}{\hbar[J + (3\langle\varepsilon\rangle/B)^{1/2}]} \quad (4.2)$$

When J is large, a number of authors^{1,17} have proposed to reduce the number of effective degrees of freedom to account for conservation laws. If this suggestion is adopted, the denominator in Eq. (4.2) should be replaced by $\hbar[J + (3\langle\varepsilon\rangle/2B)^{1/2}]$, which leads to insignificant changes in practice.

Numerical estimates of r_1 and r_2 are reported in Table I. To better judge the situation, data on benzene obtained in the present laboratory and on chlorobenzene^{27,34} have been included. The ion-induced dipole assumption is reasonable if the largest of estimates (4.1) and (4.2) is much larger than the distance that separates the released atom from the center of the phenyl ring in the equilibrium geometry of the molecular ion (i.e., about 2.5 Å for benzene and 3.1, 3.2, and 3.4 Å for chlorobenzene, bromobenzene, and iodobenzene, respectively). For the latter three substances, the quantum number $J=80$ corresponds to the most represented rotational state in our mass spectrometer source whereas $J=160$ measures the end of the Boltzmann distribution.

A study of the benzene, chlorobenzene, and bromobenzene ions carried out by Klippenstein indicates that the long-range potential (including a quadrupole term) agrees with *ab initio* calculations only when the distance r reaches values of the order of 7 or 8 Å.⁴¹

Thus, for experiments carried out in our two-sector mass spectrometer, only the dissociation of the iodobenzene ion

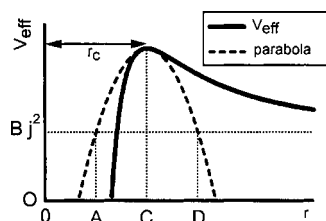


FIG. 4. Evaluation of the transit time t_{transl} [Eq. (4.3)]. Solid line: typical effective potential [Eq. (2.10)]. Dashed line: parabolic approximation to it.

approaches the conditions of validity of the assumed long-range potential, because the dynamics takes place when the length of the carbon-iodine bond is stretched away from its equilibrium position by about 2 Å or more. In the case of the bromobenzene ion, we conclude that the validity of the assumption is restricted either to experiments carried out on rotationally cold (small J) molecules, or to experiments^{35,36} involving low internal energies, i.e., long storage times and millisecond time scale measurements. The ion-induced dipole potential is unreliable for reactions leading to the release of a chlorine atom and becomes untenable when a hydrogen atom is released. Reactions involving the loss of a hydrogen atom can be meaningfully studied by OTST and SACM at very low energies or temperatures only.

B. Adiabaticity

A second problem concerns the conditions of validity of an adiabatic separation between angular and radial degrees of freedom.

The implementation of SACM has shown that it is necessary to distinguish between localized and nonlocalized nonadiabatic couplings.^{45,61,62} The former corresponds to a situation where the channel potential energy curves cross at a particular value of r . However, in the simple case where one of the fragments is an atom, the coupling is nonlocal. Following the guidelines provided by Nikitin and co-workers,^{3,63,64} we propose to define a Massey adiabatic parameter as follows.

Translational and rotational motions are separable if the adiabatic parameter ξ is much larger than 1:

$$\xi(\ell, j) = t_{\text{transl}}(\ell)/t_{\text{rot}}(j) = \omega_{\text{rot}}(j)t_{\text{transl}}(\ell) \gg 1, \quad (4.3)$$

i.e., if the translational motion is much slower than rotation.

In this inequality, $t_{\text{transl}}(\ell)$ denotes the time required to travel through a distance characteristic of the range of the effective radial potential and $t_{\text{rot}}(j) = \omega_{\text{rot}}(j)^{-1}$ is the rotational period.

The latter quantity is easily evaluated. The rotational energy E_r can be expressed as $(1/2)I\omega_{\text{rot}}^2 = Bj^2$ with the moment of inertia $I = \hbar^2/2B$. Thus, $\omega_{\text{rot}} = 2Bj/\hbar$.

The evaluation of t_{transl} is less obvious. A distance AD characterizing the range of the potential is defined in Fig. 4 where A and D are the crossing points between a parabolic approximation to V_{eff} and a horizontal line drawn at the asymptotic value Bj^2 of V_{eff} . Since, for an ion-induced dipole potential, $OA = r_c/2$ and $OD = 3r_c/2$, the characteristic distance AD is found to be precisely equal to r_c given by Eq. (2.11). The equation of motion determined by the parabolic

approximation to V_{eff} is easily derived and inverted. Standard calculations give the following estimate of the time to cross the distance AD:

$$t_{\text{transl}}(\ell) = (\mu/\ell\hbar)(\alpha q^2/\varepsilon)^{1/2}. \quad (4.4)$$

Therefore, for a given value of the quantum numbers ℓ and j , the Massey parameter is found to be equal to

$$\xi(\ell, j) = (2B\mu/\hbar^2)(\alpha q^2/\varepsilon)^{1/2}(j/\ell). \quad (4.5)$$

Adiabatic channel potential curves are meaningful only for the channels for which $\xi(\ell, j) \gg 1$. To analyze this, let us distinguish two cases.

(1) If the total angular momentum $J=0$, then $j=\ell$. The criterion for adiabatic behavior is thus, from Eq. (4.5),

$$\varepsilon \ll 4B^2\mu^2\alpha q^2/\hbar^4. \quad (4.6)$$

The right-hand side of inequality (4.6), measured in eV, is equal to 15.3, 6.3, 1.7, and 10^{-3} for reactions involving loss of I, Br, Cl, and H, respectively, when the charged partner is a phenyl ion. Clearly, there is no problem for reactions involving the loss of a heavy, strongly polarizable atom like iodine or bromine, especially if they are studied close to their reaction threshold. By contrast, SACM and OTST are again found to be reliable for the study of hydrogen loss reactions at very low energies or temperatures only.

(2) Criterion (4.6) retains some validity even when J differs from zero, because when ℓ varies from $|J-j|$ to $J+j$, the case $\ell=j$ represents the average behavior. However, Eq. (4.5) shows that nonadiabaticity can be strong when $\ell \gg j$. This will be especially the case for that part of phase space located in the neighborhood of $\ell=J+j$ line in Fig. 1. A high value of J can thus get SACM and OTST into trouble. To analyze this in more detail, let us calculate an average adiabatic parameter $\xi(E_r, J)$ by integrating Eq. (4.5) along the “pathological line” $\ell=J+j$ between 0 and $j_{\text{max}} = (E_r/B)^{1/2}$:

$$\begin{aligned} \xi(E_r, J) &= \sqrt{\frac{B}{E_r}} \int_0^{\sqrt{E_r/B}} \xi(j+J, j) dj \\ &= \left(\frac{2B\mu\sqrt{\alpha q^2}}{\hbar^2\sqrt{\varepsilon}} \right) \left[1 - \sqrt{\frac{BJ^2}{E_r}} \ln \left(1 + \sqrt{\frac{E_r}{BJ^2}} \right) \right]. \end{aligned} \quad (4.7)$$

Because the integration has been carried out along the locus of most unfavorable situations, the detailed criterion for adiabaticity, $\xi(\ell, j) \gg 1$, can be replaced by a simple inequality $\xi(E_r, J) > 1$.

So far, a subsystem including only the rotational and translational degrees of freedom has been considered. However, the rotational energy is not selected in the experiment because the subsystem is immersed in a bath of oscillators characterized by a density of states ρ_v . A last integration over E_r is required to derive a final criterion of vibrational adiabaticity:

$$\xi(J) = \frac{\int_0^{E-\varepsilon} \xi(E_r, J) \rho_v(E-\varepsilon-E_r) dE_r}{\int_0^{E-\varepsilon} \rho_v(E-\varepsilon-E_r) dE_r} > 1. \quad (4.8)$$

An analytic evaluation of Eq. (4.8) is possible if two simplifications are introduced.

(i) The density of vibrational states is assumed to increase exponentially [i.e., $\rho_v(E_v) \propto \exp(\omega E_v)$].

(ii) The upper integration limit is shifted from $E - \varepsilon$ to ∞ , both in the numerator and in the denominator.

The use of an exponential function to parametrize $\rho_v(E_v)$ instead of a power law is unusual. However, the actual increase calculated by a state-counting Beyer–Swinehart routine^{1,2} (thus taking zero-point energies into account) is found to be intermediate between a power law and an exponential function. The use of the latter would be questionable for the calculation of an absolute quantity like the rate constant. However, the parameter calculated in Eq. (4.8) is an average. Even if the weighting function is not quite appropriate, its appearance in both the numerator and the denominator reduces the inaccuracy. In addition, the validity of these approximations has been checked by numerical integrations.

The analytical result is expressed in terms of hypergeometric functions which, fortunately, can be approximated by a simple rational approximation when $0 \leq J(\omega B)^{1/2} \leq 4$. Furthermore, as said before, the experimentally determined KERDs are only significant in the energy range $[0, 3\langle\varepsilon\rangle]$. Therefore, the required adiabatic parameter has been calculated for $\varepsilon = 3\langle\varepsilon\rangle$. The final condition for adiabaticity is found to be given by

$$\xi(J) = \left(\frac{2B\mu\sqrt{\alpha q^2}}{\hbar^2\sqrt{3\langle\varepsilon\rangle}} \right) (1 + 3.2J\sqrt{\omega B} - 0.3\omega B J^2)^{-1} > 1. \quad (4.9)$$

Let us apply Eq. (4.9) to the halogenobenzene ions, using the molecular parameters reported in Sec. III. Average translational energy releases measured in a two-sector mass spectrometer have been determined by Burgers and Holmes³⁴ and by Urbain *et al.*²⁷ The vibrational density of states of the phenyl ring, $\rho_v(E_v)$, was observed to be qualitatively fitted by an exponential function of the form $C \exp(\omega E_v)$ with parameter ω equal to 10 eV^{-1} . The results, calculated for three different values of J , are reported in Table II.

When the molecules are rotationally cold, the adiabaticity criterion is seen to be fulfilled for the three reactions involving the loss of a halogen atom. For the most represented rotational state in an electron impact mass spectrometer source ($J=80$), only the bromobenzene and iodobenzene cations fulfill the criterion. At the end of the Boltzmann distribution ($J=160$), OTST can be expected to be fully reliable only for the dissociation of the iodobenzene cation.

Note that the energy $3\langle\varepsilon\rangle$ is substantially higher than the thermal energies usually considered in the applications of SACM. Note furthermore that for reactions leading to the loss of a hydrogen atom, Eq. (4.9) becomes a very stringent condition, because of the smallness of the reduced mass μ and of the polarizability α .

To summarize, we have derived two necessary conditions of validity of OTST that are both formulated in terms of the average translational energy release $\langle\varepsilon\rangle$ and not in

terms of the internal energy E . The lower the released translational energy and the total angular momentum, the larger the reduced mass, the rotational constant of the molecular fragment, and the polarizability of the released atom, the more valid is the OTST. From an experimental point of view, when KERDs are determined in a sector mass spectrometer, OTST can be expected to be somewhat reliable for the dissociation of the iodobenzene ion only, provided that the reaction is studied at internal energies as low as possible. Measurements should then preferably be carried out in the second field-free region of a reverse geometry instrument.

For these reasons, the conclusions reached by Lim, Choe, and Kim,⁴⁰ who applied phase-space theory to the dissociation of the bromobenzene ion on a nanosecond time scale, sometimes considering internal energies as high as 10 eV seem to us unreliable. The inadequacy of phase space theory or OTST at too high energies may explain why these theories are unable to provide an adequate fit to the experimental data unless the reaction critical energy is taken as an adjustable parameter.⁴⁰

V. THE MAXIMUM ENTROPY METHOD

A. Entropy deficiency and phase-space sampling

Our second tool in the analysis of KERDs is the maximum entropy method, which is derived from the information theory approach to statistical mechanics. Its aim is not to predict quantitatively a KERD, but rather to analyze experimental data in order to assess the efficiency of phase-space sampling and to identify the dynamical constraints that prevent ergodicity.^{19–25} Our purpose is to apply it to the analysis of the dynamical constraints that derive from angular momentum conservation. The method is very general: unlike OTST, its validity is not limited to a certain range of energy parameters.

Basically, the method starts from the so-called “prior distribution” $P^0(\varepsilon|E)$, where at a given internal energy E , all quantum states are assumed to be equally probable, whatever their combination of vibrational, rotational, or orbital quantum numbers be:

$$P^0(\varepsilon|E) \propto \varepsilon^{1/2} \rho_{\text{vr}}(E - \varepsilon). \quad (5.1)$$

In this equation, $\varepsilon^{1/2}$ is the energy dependence of the density of states corresponding to the three-dimensional relative translational motion of the fragments. The second factor represents the density ρ_{vr} of the remaining rotational-vibrational degrees of freedom that accept the complementary energy $E - \varepsilon$.

The prior distribution $P^0(\varepsilon|E)$ is derived independently of any dynamical properties related to the interaction potential. It therefore represents the fully statistical distribution.

It then can be shown^{19–24} that the least biased way to convert the prior to an actual distribution is to submit it to a sequence of multiplicative corrections

$$P(\varepsilon|E) \propto P^0(\varepsilon|E) \exp[-\lambda_0(E)] \prod_{r=1}^n \exp[-\lambda_r(E) A_r(\varepsilon)], \quad (5.2)$$

where the quantities A_r are appropriate physical observables whereas λ_r are the conjugated Lagrange parameters. The average values $\langle A_r \rangle$ of the observables A_r over the actual distribution $P(\varepsilon|E)$ are denoted as the constraints. Therefore, the first corrective factor $\exp[-\lambda_0(E)]$ is chosen so that the distribution is normalized in the following way:

$$\int_0^E P(\varepsilon|E) d\varepsilon = \int_0^E P^0(\varepsilon|E) d\varepsilon = 1. \quad (5.3)$$

The function $P(\varepsilon|E)$ defined in Eq. (5.2) is the only distribution that generates average values $\langle A_r \rangle$ equal to the constraints and that, at the same time, is characterized by a maximum entropy.

Furthermore, it can be demonstrated that a sequence of multiplicative corrections of the form (5.2) converges to an exact distribution.⁶⁵ However, the power of the method lies in the fact that a small number of constraints (usually just one for barrierless reactions) suffices to convert the prior distribution into an approximation comparable to the experimental result.

Assume therefore that the dynamics is determined by just a single constraint, i.e., the index n appearing in Eq. (5.2) can be taken equal to one. It then follows that a straight line is obtained when the logarithm of the ratio between the prior distribution and an actual KERD is plotted as a function of A_1 . This plot, which is termed a surprisal graph, provides a way to identify the constraint that operates on the dynamics.^{19–23}

The discrepancy between the prior and the experimental distributions can be quantitatively analyzed via the calculation of entropy deficiencies. Each distribution is characterized by its entropy and the prior distribution P^0 has the highest possible entropy S_0 . The objective is to compare them in order to derive information on the dynamical (i.e., nonstatistical) effects that operate. The difference between the entropies of the prior distribution $P^0(\varepsilon|E)$ and of an experimental KERD $P_1(\varepsilon|E)$ is termed as the entropy deficiency DS_1 , which can be shown to be equal to

$$DS_1(E) = S_0(E) - S_1(E) = \int_0^E P_1(\varepsilon|E) \ln \left[\frac{P_1(\varepsilon|E)}{P^0(\varepsilon|E)} \right] d\varepsilon \geq 0. \quad (5.4)$$

A nonzero value for DS implies that the pair of fragments does not sample the whole of the phase space that is compatible with the mere specification of the internal energy. Furthermore, the quantity $\exp(-DS)$ has been shown to be an upper bound to the ratio between the volumes of phase space effectively sampled by the two distributions.^{24–30} Therefore, it can be referred to as an ergodicity index.

B. Comparison between two distributions

Consider now two constrained distributions, say $P_1(\varepsilon|E)$ and $P_2(\varepsilon|E)$, both of which are derived from the prior $P^0(\varepsilon|E)$ by the action of different constraints. Their entropy deficiencies are linked by the following relationship:

$$DS_2 = DS_1 + \int_0^E P_2(\varepsilon|E) \ln \left[\frac{P_2(\varepsilon|E)}{P_1(\varepsilon|E)} \right] d\varepsilon + \int_0^E [P_2(\varepsilon|E) - P_1(\varepsilon|E)] \ln \left[\frac{P_1(\varepsilon|E)}{P^0(\varepsilon|E)} \right] d\varepsilon. \quad (5.5)$$

The quantity

$$H_{21} = \int_0^E P_2(\varepsilon|E) \ln \left[\frac{P_2(\varepsilon|E)}{P_1(\varepsilon|E)} \right] d\varepsilon \quad (5.6)$$

measures the discrepancy between two distributions but is not an entropy deficiency. It has been termed the persuasion index²⁰ on the following grounds. If $P_2(\varepsilon)$ denotes the experimental KERD whereas $P_1(\varepsilon)$ represents the distribution predicted by a particular theory, like OTST, then a small value of H_{21} indicates a high reliability of that theory. The OTST modelization is satisfactory if H_{21} is smaller than the usual χ^2 measure of the experimental error:

$$H_{21} < (1/2) \int_0^E P_2(\varepsilon) \left[\frac{\delta P_2(\varepsilon)}{P_2(\varepsilon)} \right]^2 d\varepsilon, \quad (5.7)$$

where $\delta P_2(\varepsilon)$ is the uncertainty in the experimental KERD.

Equations (5.4)–(5.7) will be used to appraise the reliability of OTST in Sec. VI.

C. Two- and three-dimensional prior distributions

If the central field approximation is valid, then the nuclear trajectories that generate the KERD are constrained to be planar. The density of translational states of a particle moving in a two-dimensional (2D) box is a constant. Therefore, when OTST is valid, the appropriate prior distribution $P_{2D}^0(\varepsilon|E)$ is simply proportional to $\rho_{vr}(E-\varepsilon)$, and thus is a steadily decreasing function of ε :

$$P_{2D}^0(\varepsilon|E) \propto \rho_{vr}(E-\varepsilon). \quad (5.8)$$

The two-dimensional prior distribution should be distinguished from the more general three-dimensional prior defined earlier in Eq. (5.1), which is a reference under all circumstances and which henceforth will be denoted as $P_{3D}^0(\varepsilon|E)$.

The use of Eq. (5.8) as a prior distribution that satisfies the conservation of angular momentum has been advocated by Quack and Troe.⁶⁶ Note also that a two-dimensional Boltzmann distribution has previously been introduced in the study of KERDs by Klots^{60,67} on the analogy of thermal processes.

VI. RESULTS

As shown in Fig. 5, $\tilde{P}_{\text{OTST}}(\varepsilon)$ (where the tilde denotes an average over the thermal distribution of J and over the experimental distribution of internal energies E) is intermediate

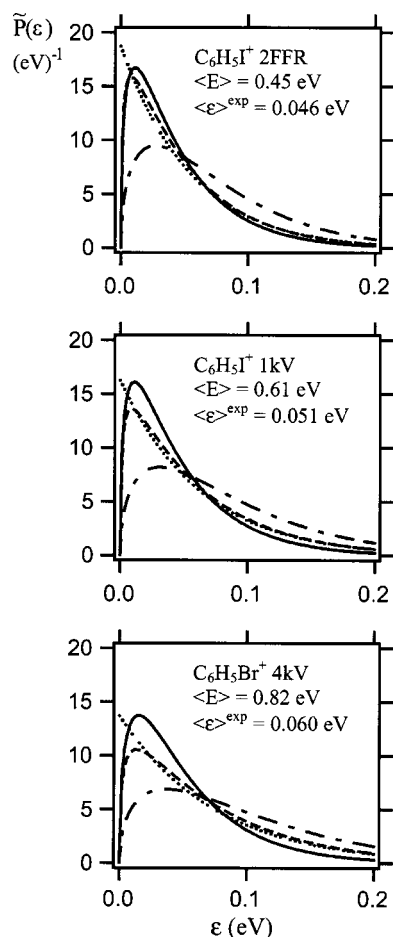


FIG. 5. KERDs for the dissociations of the iodobenzene and bromobenzene cations. Solid line, experiment; dashes, OTST calculations; dots, 2D prior distribution; dot-dashed line, 3D prior distribution.

between the prior $\tilde{P}_{3D}^0(\epsilon)$ and $\tilde{P}_{exp}(\epsilon)$. The maximum entropy method can provide a quantitative measure of the observed discrepancies. The persuasion index $H(\text{exp-OTST})$ as well as various entropy deficiencies have been calculated from Eqs. (5.4) and (5.6) and are gathered in Table III.

A. Range of validity of OTST

The introduction of the experimental error bars into Eq. (5.7) generates a criterion for the validity of OTST: the modulation of the experimental KERD is satisfactory if the per-

suation index H is lower than 0.01. Only the low-energy reaction where $\text{C}_6\text{H}_5\text{I}^+$ dissociates in the second field-free region ($\langle E \rangle = 0.45$ eV) fulfills this criterion ($H = 3 \times 10^{-3}$). As clearly shown in Table III and in Fig. 5, the higher the internal energy E , the less reliable is the OTST.

B. Relevant constraints

In all cases reported in Table III, the entropy deficiencies between the experimental KERDs and the three-dimensional prior P_{3D}^0 , i.e., $\text{DS}(\text{exp-3D}) = S_{exp} - S_{3D}^0$ are found to be much larger than those measured with respect to P_{2D}^0 , i.e., $\text{DS}(\text{exp-2D}) = S_{exp} - S_{2D}^0$. This strongly suggests that restricting the dynamics to a two-dimensional space provides a much better starting point than a 3D description. In other words, the major constraint that affects P_{exp} is the requirement that the translational motion be two dimensional, thereby quantitatively confirming the visual impression given by Fig. 5.

The same constraint acts on the OTST distribution owing to the central field assumption. Surprisal graphs have been calculated for P_{OTST} when the prior distribution is the three-dimensional one, i.e., $P_{3D}^0(\epsilon)$. They are observed to be well described by the following equation:

$$I = \ln(P^0/P) = \lambda_0 + \lambda_1 \ln(\epsilon^{1/2}) \quad (6.1)$$

with $\lambda_1 = 1 \pm 0.1$. This unexpected behavior indicates that the multiplicative correction $\exp[-\lambda_1 A_1(\epsilon)]$ of Eq. (5.2) is proportional to $\epsilon^{-1/2}$, thus exactly canceling the three-dimensional translational density of states appearing in Eq. (5.1).

Note that $\text{DS}(\text{exp-2D})$ remains small even at high internal energies where the central field approximation breaks down. However, since it is not equal to zero, even when $H(\text{exp-OTST})$ is negligible, additional constraints are clearly seen to operate on P_{2D}^0 whatever the sampled internal energy. They are responsible for the threshold behavior of the KERD.

The nature of these additional constraints is clear at low internal energies, when OTST reproduces the experiment. When the molecular fragments are endowed with an internal structure, constraints concerning the relative magnitude of j and ℓ [Eq. (2.5a)] and the requirement that the translational energy be large enough to surmount the orbital barrier [Eq. (2.5b)] show up. They determine the magnitude of $\text{DS}(\text{exp-}$

TABLE III. Persuasion index H and entropy deficiencies DS . $\langle E \rangle$ is the average internal energy, $\langle \epsilon \rangle$ is the measured average translational energy released, and $\langle J \rangle$ is the average total angular momentum. The second column indicates the fragment ion translational energy (in the laboratory frame) for spectra recorded in the first field-free region of the spectrometer; 2FFR indicates that the measurement has been done in the second field-free region at an accelerating voltage of 8 kV.

Ion	Scan	$\langle E \rangle$ (eV)	$\langle \epsilon \rangle$ (eV)	$\langle J \rangle$	H (exp-OTST)	DS (exp-3D)	DS (exp-2D)	DS (OTST-2D)
$\text{C}_6\text{H}_5\text{I}^+$	2FFR	0.45	0.046	89	0.003	0.15	0.03	0.02
$\text{C}_6\text{H}_5\text{I}^+$	1 keV	0.56	0.045	89	0.02	0.24	0.05	0.02
$\text{C}_6\text{H}_5\text{I}^+$	3 keV	0.61	0.051	89	0.02	0.21	0.04	0.02
$\text{C}_6\text{H}_5\text{Br}^+$	2FFR	0.65	0.059	81	0.02	0.16	0.03	0.02
$\text{C}_6\text{H}_5\text{Br}^+$	1 keV	0.73	0.060	81	0.03	0.21	0.04	0.02
$\text{C}_6\text{H}_5\text{Br}^+$	4 keV	0.82	0.060	81	0.04	0.26	0.05	0.02

2D) and are found to be much less restrictive than the major constraint. However, at higher internal energies, the OTST assumptions become unreliable. The corresponding constraints are inadequate and should be replaced by others whose interpretation is not obvious.

The choice of $P_{2D}^0(\varepsilon|E)$ has been advocated by Quack and Troe⁶⁶ in order to account for conservation of angular momentum. It is then necessary to include a further correction due to threshold effects. However, this correction is slight. In addition, Nesbet and Clary^{68,69} found by quantum dynamical calculations that the threshold law is proportional to $\varepsilon^{1/2}$ as in the three-dimensional prior distribution or in OTST.

C. OTST is not constrained enough

Note also that at all energies, the entropy deficiencies predicted by OTST are too small: $DS(\text{exp-2D})$ is always larger than $DS(\text{OTST-2D})$. Thus, $P_{\text{OTST}}(\varepsilon)$ is less constrained than the experimentally observed KERD and the active part of phase space is actually smaller than that corresponding to the subspace Ω defined by Eqs. (2.5) and represented in Fig. 1. As a matter of fact, OTST is known to predict rate constants and reaction cross sections that are invariably too large.^{1,9,10,13,45,46,61,64,70} This implies that the actual bottleneck is more stringent than that calculated by OTST.

VII. DISCUSSION

A. Flaws in OTST

The right way to mend OTST is not at all obvious. Several lines of thoughts can be envisaged.

The harmonic approximation in the calculation of energy-level densities is *a priori* questionable. The use of anharmonic oscillators would increase the density of vibrational states. Therefore, more energy would flow into the bath of oscillators and less into the reaction coordinate. The average translational energy release that is experimentally observed is indeed less than that predicted by the OTST distribution. However, the corrections to the density of states that result from anharmonicity are known to decrease with the number of degrees of freedom. Detailed calculations have been carried out by Forst.⁴ At internal energies of the order of 2.5 eV, the ratio $\rho_{\text{anh}}/\rho_{\text{harm}}$ of the densities of states is less than 1.2 for triatomic molecules and less than 1.1 for tetraatomic systems. At the internal energies considered here (i.e., less than 3.6 eV), anharmonicity is expected to have a negligible influence on the density of states of the phenyl ion (27 oscillators) because the mean energy per vibrational mode is of the order of 1000 cm^{-1} .

An additional approximation has been introduced in order to calculate the rotational-orbital density of states: the phenyl ion has been treated as a spherical top. A closed-form expression for the rotational levels of an asymmetric top is unknown. However, the position of these levels can be obtained by interpolation between a prolate and an oblate symmetric top. Thus, a comparison between the pattern of rotational levels of spherical, symmetric, and asymmetric tops reveals essentially degeneracies for the former that are lifted for the latter. Since the function that represents the density of

states is a smooth average over degeneracies, the use of a spherical top approximation should be of no consequence. As a matter of fact, the rotational density of states depends only on the product of the moments of inertia. Indeed, the usual practice is to modelize the molecule as a collection of individual one- or two-dimensional rotors, irrespective of the nature of the top.

The reliability of OTST is restricted to low internal energies because it depends on the validity of Eqs. (2.5) and (2.10). OTST has been found in Sec. VIC to be not constrained enough to properly account for experimental results obtained at too high energies. As a matter of fact, attempts to improve on it can make it even less constrained. As shown by Eq. (2.5a), OTST is based on a vectorial coupling scheme between angular momenta of rotors and orbital motion that is valid only when the rotors remain rigid and when the vibrational degrees of freedom undergo motions with vanishingly small amplitudes.⁷¹ As a consequence, the higher the internal energy E , the less relevant are the inequalities (2.5a) to the control of phase-space sampling.

The same is true for inequalities (2.5b) or (2.5c). The genuine interaction potential can be expected to contain higher-order terms in its multipolar expansion, and thus can be expected to have a steeper slope than that of the pure ion-induced dipole interaction. This implies that values of ℓ larger than the quantity ℓ_{max} predicted by the Langevin model [Eq. (2.5b) or Eq. (2.5c)] should be accepted in the active space Ω , which has been found to be already too large.

Restriction to phase-space sampling is obtained, however, if the central field approximation is abandoned. No torque can arise between two spherical or dimensionless partners linked by an isotropic interaction. By contrast, constraints resulting from conservation of angular momentum might be stringent if the dissociation generates an ion and a strongly polar neutral fragment.

Even if the neutral fragment is nonpolar, the central field approximation breaks down at high internal energies because the ion-induced dipole potential is contaminated by anisotropic short-range valence contributions. Anisotropic potentials have been thoroughly studied by Troe and co-workers on the example of the ion-permanent dipole and ion-quadrupole potentials. Anisotropy was shown to introduce in the dynamics a constraint that reduces the amount of phase space actually sampled.^{45,46,61,64} Its influence can be accounted for by multiplying the results provided by phase-space theory by a corrective factor f_{rigid} denoted “rigidity factor,” which is less than unity. In other words, the rigidity factor considered in SACM plays a similar role as the entropy deficiency in the maximum entropy method. Both concepts account for the fact that phase-space theory always overestimates rate constants.

B. Number of relevant degrees of freedom: translation and rotation

The maximum entropy method analyzes the situation in more general terms than does OTST. It has no knowledge of the reliability of the estimate of the barrier and ignores whether the $\{j, \ell\} \rightarrow \vec{J}$ coupling scheme is relevant. From a

comparison between the experimental KERD and a reference prior distribution, either three- or two-dimensional, it detects nonzero entropy deficiencies, i.e., incomplete phase-space sampling. This information is reliable even when OTST is invalid. If the experimental KERD is compared with a 3D prior, then the first—and major—correcting factor to be applied in Eq. (5.2) consists in reducing the number of translational degrees of freedom, i.e., in switching to a 2D distribution. If the experimental KERD is now compared with a 2D prior, then the next correcting factor truncates the low-energy part of the distribution. The interpretation is straightforward when OTST is valid, but it is very surprising to find a hint that the dimensionality reduction persists as a propensity at higher energies when the ion-induced dipole potential [Eq. (2.10)] is no longer reliable. A possible explanation can be put forward.

The calculation of a KERD is based on the assumption that the dynamics remain vibrationally adiabatic as the fragments separate in space, i.e., along the last part of the reaction coordinate. This is usually the case for the so-called conserved modes (i.e., those that change very little during the reaction process). However, the adiabatic parameter derived in Sec. IV B is related to transitional modes, i.e., those that correlate with the overall rotations of the fragments and with the orbital motion. These modes play an essential role in the conservation of angular momentum, in the generation of a centrifugal barrier, and in its conversion into translational energy. Unfortunately, they are strongly anharmonic and nonseparable, and their study raises very difficult problems.^{62,72}

It has been seen repeatedly that a major condition of validity of OTST is a large value of r_c , i.e., a low value of ℓ . This requires a low excitation of the transitional modes. Therefore, OTST is valid not only immediately close to the reaction threshold, but also when the excess energy is released as purely vibrational energy stored in conserved modes. The conserved modes may be thought of as constituting a vibrational bath whereas the transitional modes form a subsystem. For reaction (3.1) studied here, the size of the bath (27 for the C_6H_5^+ fragment) is much larger than that of the subsystem, which reduces to the two CX rocking motions. Hence, most of the energy flows into the bath, leaving the remainder relatively unexcited and thus unable to push the rotational barrier down to short interfragment separations. More generally, it may be concluded that the larger the molecule, the more numerous are the conserved modes; the smaller the ratio between the number of transitional and conserved modes, the lower is the average excitation of the transitional modes, and more valid is the OTST or, at least, more plausible is the assumption that the dynamics takes place in an approximately two-dimensional space. However, the other assumptions of OTST may become unreliable as the size of the molecule increases: the dissociation of a large molecule usually generates floppy fragments interacting via a complicated potential function.

A reduction of the number of rotational degrees of freedom has also been proposed a long time ago,^{1,17} at least in the low- J limit. Consideration of Fig. 2 explains why the low- and high- J cases should be treated differently. The high-

energy part of the KERD (approximately from the maximum onwards) is determined by the upper part of Fig. 2, i.e., by that part of the $(j_{\max}, \ell_{\max})$ plane above the line $\ell_{\max} = J + j_{\max}$. It appears from Fig. 2 that for the sphere-atom interaction ($\nu=1$), the rotational-orbital density of states ρ_{ro} is proportional to $E_r^{1/2}$ in the high- J case, which implies participation of three rotational degrees of freedom. On the other hand, ρ_{ro} is independent of E_r in the low- J case, which implies that the number of rotational degrees of freedom has been reduced to two. A similar analysis applies to the interaction between a linear molecule and an atom ($\nu=0$).

C. Possible deeper reasons

Reasons of a more fundamental nature can be envisaged to account for the persistence at higher energy of the dimensionality reduction. Two research directions may be proposed to investigate a possible approximate conservation of the orbital angular momentum of the relative motion independently of the internal motion of the core.

(1) The Kolmogorov–Arnold–Moser (KAM) theorem⁷³ can be invoked. Its conditions of validity have been shown to be fulfilled for some particular central field problems.⁷⁴ In our case, the unperturbed Hamiltonian derives from the isotropic Langevin potential and the perturbation sets in as the top of the centrifugal barrier moves to short internuclear separations. If the internal energy is low enough, i.e., if r_c is large enough [Eq. (2.13)], trajectories can be expected to remain confined to a restricted subspace of phase space. Note in this connection that orbiting motions have been demonstrated by Pechukas to exist even in noncentral force fields.⁷⁵ This opens the possibility of using angle-action variables to study the problem.

(2) At high internal energy, the central field approximation breaks down and the reaction coordinate is characterized by its curvature. The transitional modes can, to a first approximation, be reasonably assumed to behave as oscillators orthogonal to the reaction coordinate. Miller, Handy, and Adams have developed a reaction path Hamiltonian method⁷⁶ that provides a way to study the reaction dynamics in that case. Cole and Balint-Kurti⁷⁷ have recently implemented a promising “statistical adiabatic product distribution method,” which consists of a reaction path calculation of KERDs that takes account of the conservation of angular momentum. Unfortunately, a realistic *ab initio* calculation of the evolution of the transitional modes along the reaction path is fraught with great difficulties and requires top-ranking computational methods.

A particularly interesting possibility is that the conservation properties of ℓ , which derive from the central field approximation, should, when the energy increases, be replaced by adiabatic invariance. Miller and co-workers^{76,78,79} could show that when the curvature is not too large, it is possible to separate adiabatically the reaction path and the oscillators perpendicular to it. Adiabatic invariance introduces a constraint in the dynamics that should be detectable by the maximum entropy method. However, when J is nonzero, the angular momentum and the coupling coefficients that result from curvature are intricately intermeddled. As a conse-

quence, it is no longer possible to sort out which part of the entropy deficiency results from the curvature of the reaction path and which part is due to constraints related to angular momentum conservation.

VIII. CONCLUDING REMARKS

A. Two-dimensional dynamics

The joint theoretical and experimental analysis presented in this work leads to the conclusion that, even when the central field assumption is questionable, the major constraint that results from conservation of angular momentum is linked with the reduction of the dimensionality of the translational motion, from 3D down to 2D. A second constraint, deriving from the $\{\vec{j}, \vec{\ell}\} \rightarrow \vec{J}$ coupling scheme, leads to a suppression of small translational releases and accounts for threshold behavior. Its influence on the entropy deficiency is less important. At high internal energies, the influence of conservation of angular momentum cannot be separated from effects resulting from curvature of the reaction path.

B. A statistical view

The maximum entropy method is not a predictive but an interpretative theory that provides a valid tool at all internal energies. The derived information is less detailed, but retains its reliability even when predictive theories break down.

The reasons of the success of the maximum entropy method can be understood as follows. Consider the OTST procedure. The connection between a set of theoretical conditions concerning angular momentum conservation [Eqs. (2.5)] and an experimentally observable quantity (the KERD), requires a sequential procedure. A rotational-orbital density of states ρ_{ro} has to be determined first by integration over an active part of phase space determined by Eqs. (2.5). The function ρ_{ro} is then convoluted with a vibrational density of states to generate a J -resolved KERD, denoted by $P(\varepsilon|E, J)$. Next, the latter function has to be averaged over J to give the energy-resolved KERD $P(\varepsilon|E)$. This function must then be further integrated over the parent ion internal energy $T(E)$ to give the actually observable KERD, $\tilde{P}(\varepsilon)$. Note also that, most of the time, the experimentalist prefers to argue in terms of an average kinetic energy release $\langle \varepsilon \rangle = \int \varepsilon \tilde{P}(\varepsilon) d\varepsilon$, or in terms of a rate constant $k(E)$, which is also related to the integral of the KERD. Thus, altogether, the connection between the angular momentum conservation laws and experiment requires a sequence of four or five integration, convolution, or averaging steps. No doubt, this is an ideal situation for a statistical theory.

Obviously, going the same way in reverse order is impossible. However, what the maximum entropy method attempts to do is less ambitious. It does not try to go back from experiment to a specification of the laws that delineate the active part of phase space, e.g., Eqs. (2.5). Its aim is merely to relate the averaged experimental result to the magnitude of the active subspace Ω , which is another average quantity.

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APPENDIX A: EVALUATION OF ρ_{ro}

To evaluate Eq. (2.8), the variables j and ℓ are replaced by

$$x = 2^{-1/2}(j - \ell), \quad (\text{A1a})$$

$$y = 2^{-1/2}(j + \ell - J). \quad (\text{A1b})$$

Since the Jacobian is equal to 1, Eq. (2.8) becomes

$$\rho_{ro}(E_r, \varepsilon, J) = \frac{\sqrt{2}}{B} \int_{\Omega} dx dy [\sqrt{2}(x + y) + J]^{\nu-1} \times \left| y - \left(\sqrt{2}j_{\max} - x - \frac{J}{\sqrt{2}} \right) \right|. \quad (\text{A2})$$

The variable ε is contained in the integration range Ω in the form of $\ell_{\max}$ via Eq. (2.12), while $E_r = Bj_{\max}^2$.

Let us, for example, evaluate integral (A2) when $j_{\max} > J$ and when $\ell_{\max} > j_{\max} + J$ (Fig. 3). The variable x ranges between $-J/\sqrt{2}$ and $+J/\sqrt{2}$, while y extends between 0 and the straight line $j = j_{\max}$, i.e., up to $2^{1/2}j_{\max} - 2^{-1/2}J - x$. Integrating Eq. (A2) in this subspace leads to

$$\rho_{ro}(E_r, \varepsilon, J) = \left(\frac{2J}{B} \right) (2j_{\max})^{\nu-1}.$$

Altogether, nine different cases can be recognized. However, some of them lead to identical results. The overall situation is summarized in Fig. 2.

APPENDIX B: THRESHOLD LAW

When ε is very small, so is $\ell_{\max}$ as a result of Eq. (2.5c). The rotational-orbital density of states ρ_{ro} is nonzero in the range $j_{\max} \in [J - \ell_{\max}, J + \ell_{\max}]$ and is then equal to (Fig. 2)

$$\rho_{ro} = (\ell_{\max} - |J - j_{\max}|)(2j_{\max})^{\nu-1}/B. \quad (\text{B1})$$

The threshold law is derived from Eq. (2.7) by substituting $j_{\max}$ in place of E_r as the integration variable and using Eq. (B1):

$$\lim_{\varepsilon \rightarrow 0} P_{\text{OTST}}(\varepsilon|E, J) \propto \int_{J-\ell_{\max}}^{J+\ell_{\max}} \rho_{\text{vib}}(E - \varepsilon - Bj_{\max}^2) \times (\ell_{\max} - |J - j_{\max}|) j_{\max}^{\nu} dj_{\max}. \quad (\text{B2})$$

The integration range is now split into two parts ($j_{\max} < J$ and $j_{\max} > J$) and ρ_{vib} is expanded as a Taylor series about $j_{\max} = J$. One gets

$$\lim_{\varepsilon \rightarrow 0} P_{\text{OTST}}(\varepsilon|E, J) \propto \ell_{\max}^2 + O(\ell_{\max}^4) + \dots \quad (\text{B3})$$

In fact, the zero-order term $\rho_{\text{vib}}(E - \varepsilon - BJ^2)$ suffices to get Eq. (2.9).

Note that both the maximum entropy model and OTST predict for the threshold law the same $\varepsilon^{1/2}$ dependence. However, in the former approach, the $\varepsilon^{1/2}$ factor derives from the density of states of a three-dimensional translational motion [Eqs. (5.1) and (5.2)]. In OTST [Eq. (2.14)], it results from the impossibility to surmount a rotational barrier if the energy stored in a radial motion is too low [Eq. (2.5b)]. In view of the limitations of OTST, one might be tempted to prefer the explanation in terms of the translational density of states, because it is more general. However, it remains true that when $\varepsilon \rightarrow 0$, r_c is large enough to be located in a region of space where the force field is central. Then ℓ is a good quantum number, the $\{\vec{j}, \vec{\ell}\} \rightarrow \vec{J}$ coupling scheme is reliable, and OTST is valid.

APPENDIX C: HIGH-ENERGY BEHAVIOR OF THE KERD

If ε is large enough for $\ell_{\max}(\varepsilon)$ to be greater than $J + (E_r/B)^{1/2}$ then, as shown in Fig. 2, the rotational-orbital density of states is independent of ε and can be denoted as $\rho_{ro}(E_r, J)$. Applying the first mean-value theorem⁸⁰ to Eq. (2.7), one obtains

$$P_{\text{OTST}}(\varepsilon|E, J) = N(E, J) \rho_{ro}(\bar{E}, J) \int_0^{E-\varepsilon} \rho_{\text{vib}}(E - \varepsilon - E_r) dE_r. \quad (\text{C1})$$

Expand the vibrational density of states as a power series

$$\rho_{\text{vib}}(x) = \sum_j c_j x^j \quad (\text{C2})$$

and denote the corresponding sum of states as $G_{\text{vib}}(x)$:

$$G_{\text{vib}}(x) = \int \rho_{\text{vib}}(x) dx = \sum_j \frac{c_j}{j+1} x^{j+1}. \quad (\text{C3})$$

Then,

$$\begin{aligned} P_{\text{OTST}}(\varepsilon|E, J) &= N'(E, J) \sum_j c_j \int_0^{E-\varepsilon} (E - \varepsilon - E_r)^j dE_r \\ &= N'(E, J) \sum_j c_j \frac{(E - \varepsilon)^{j+1}}{(j+1)} \\ &= N'(E, J) G_{\text{vib}}(E - \varepsilon). \end{aligned} \quad (\text{C4})$$

The factor $N'(E, J)$ is determined by normalizing the distribution according to Eq. (5.3).

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